Word Count: 2977

Roller Support Design to Prevent Smudging for Left-Handed Writing in the EngSci Common Room

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1 Introduction

This design report presents the opportunity of preventing smudging on whiteboards for left-handed EngSci students and recommends a roller ball support design based on the evaluation criteria. Currently, left-handed students must position their hands in unergonomic positions to prevent smudging the marker ink [Figure 2] [Appendix B]. Four potential solutions were created to solve this problem and after verification through testing, Pairwise Comparison Matrices, and Pugh Charts, the roller support design was determined to meet stakeholder needs the best.

2 The Problem With Left-Handed Writing on Whiteboards

English is the primary language of instruction at UofT [1]. Since this language is written from left to right, and a left handed student's hand will follow from the left of the marker writing, their hand will contact the freshly applied ink of what was just written. This causes ink on the whiteboard to smudge and transfers ink residue to the hand, as shown in Figure 1. To mitigate smudging these students often position their wrists in awkward angles while writing, as shown in Figure 2 [Appendix B]. This causes greater wrist flexion, increasing muscle strain and potentially leading to musculoskeletal disease [2], [3], [4]. Therefore, solutions are needed to prevent smudging while allowing left-handed writers to position their hands ergonomically during writing.

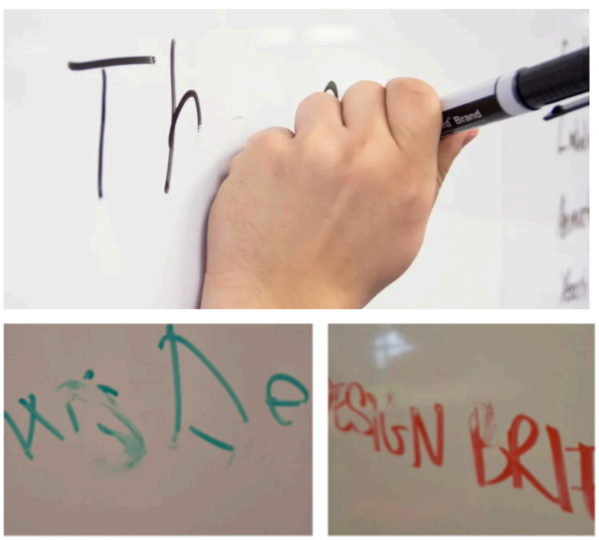

Figure 1: Examples of smudging that occurs due to left-handed writing. Text can become illegible due to smudging, reducing communicability for left-handed writers through whiteboards.

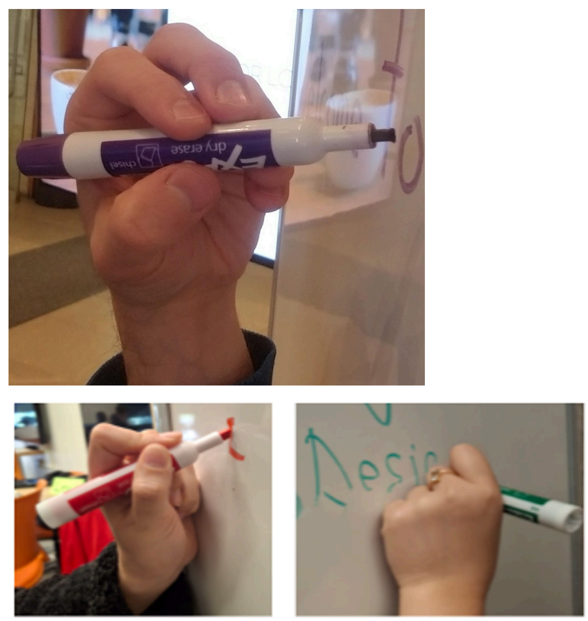

Figure 2: Multiple examples of left-handed writers straining their wrists in awkward angles to write without smudging.

3 Defining the Working Environment

The working environment is the EngSci common room, an indoor space with several porcelain whiteboards and whiteboard markers. Although UofT has not confirmed if the whiteboards are porcelain, they are likely to be since they are magnetic, non-porous, and glossy [6], [7]. Expo Dry Erase markers [8] are the prevalent marker in the common room, so designs should function properly using them. They have a maximum diameter of 2.29 cm and a length of 14.27 cm [9].

4 Stakeholders

This section lists the stakeholders for our identified opportunity and explains each stakeholder's main wants.

4.1 Left-Handed EngSci Students

Primary Stakeholders

These students want to be able to write on whiteboards at the same rate as right-handed peers without unintentionally smudging the ink on the board [Appendix B1]. They wish for the design to be ergonomic and not hinder their writing ability [Appendix B2].

4.2 Right-Handed EngSci Students Writing Leftward Scripts

Primary Stakeholders

These students face the same problem as left-handed students when writing right-to-left languages like Arabic or Hebrew [Appendix B3]. Therefore, designs should accommodate both right and left-handed users.

4.3 Audiences of Left-Handed Students

Secondary Stakeholders

The audience wants to read the writing of left-handed writers, requiring that the writing be legible with no visible smudges. They also wish for left-handed writers to write at the same rate as their right-handed counterparts.

4.4 UofT Facilities

Tertiary Stakeholders

The facilities want no permanent residue or damage done to the whiteboard. There must not be multiple maintenance required per day and standards for educational spaces of University of Toronto [10] must be followed.

5 Requirements and Evaluation Criteria

This section lists critical goals with their associated requirements and evaluation criteria that will be used to verify and compare different designs.

Need: Left-handed students need to write ergonomically on whiteboards in the EngSci common room without smudging or hindering their normal writing ability.

Table 1: Goal and associated requirement to meet stakeholder wants of maintaining writing speeds on whiteboards.

	Description	Justification
Goal 1	Design must not affect the user's baseline writing ability	
Requirement 1	Cannot inhibit the user's baseline DASH-2 score [11] by more than 15 points.	A 15 point DASH-2 score reduction is one standard deviation [12], signifying a significant reduction in writing speed
Evaluation Criteria 1	The smaller the point reduction on the DASH-2 test, the better.	

Table 2: Goal to address stakeholder wants of minimizing smudging

	Description	Justification
Goal 2:	Ensure minimal unintentional movement of the ink on the surface of the whiteboard	
Requirement 2	Contact with marker ink cannot cause writing to have a ΔE^* (Euclidean Difference) of more than 3 compared to the original ink color [24].	According to the NBS, a ΔE^* of 3 is a perceivable difference so anything lower is acceptable [24].
Evaluation Criteria 2:	The smaller the ΔE^* between unsmudged and smudged ink, the better.	

Table 3: Goal and associated requirements and evaluation criteria to ensure stakeholder ergonomics and comfort.

	Description	Justification
Goal 3	Design must ensure minimal discomfort and strain to be ergonomic	
Requirement 3.1	No wrist extension or flexion > 15^{o}, and no ulnar or radial deviations of the wrist > 15^{o} [Appendix D], [22].	The RULA Assessment is a score based system where the higher the score, the worse the risk of strain. Any deviations of the ulnar or radial bones > 15^{o} is +1 to the score and any wrist extension or flexion > 15^{o} is +2 [22].
Evaluation Criteria 3.1	The smaller the wrist deviations from neutral during use the better.	
Requirement 3.2	The weight of the design should be less than 400g including the original marker.	The Canadian Centre for Occupational Health and Safety recommends a maximum of 400g for one-handed tools to reduce the risk of developing

		musculoskeletal disorders [13].
Evaluation Criteria 3.2	The lighter the design the better.	

6 Recommended Roller Support Design

This section will describe the recommended design, justify key design decisions about it, and provide an overview of why it is the recommended design.

6.1 Roller Support Design Description

The design that we converged onto was a ball support on a wrist cuff, as shown in Figure 3. Users hold a marker normally in their hand and their wrist/arm is supported by the platform attached to the ball contacting the whiteboard. The ball does not smudge noticeable amounts of ink on the whiteboard because if it doesn't slip, it only pushes ink into the whiteboard. [Video].

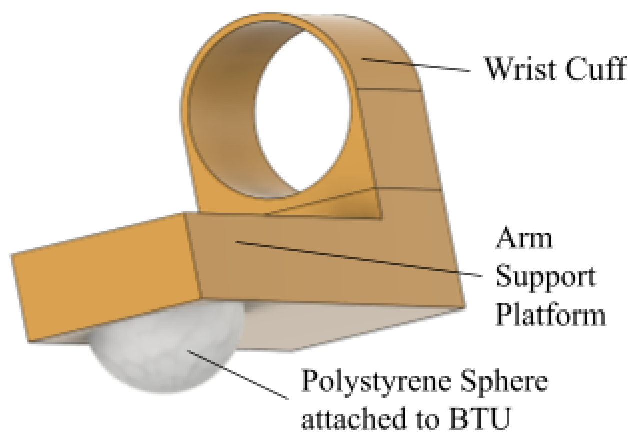

Figure 3: Roller support design. The arm rests on the flat platform and the wrist goes through the wrist cuff. The sphere is attached to the ball transfer unit (BTU) as shown in Figure 4.

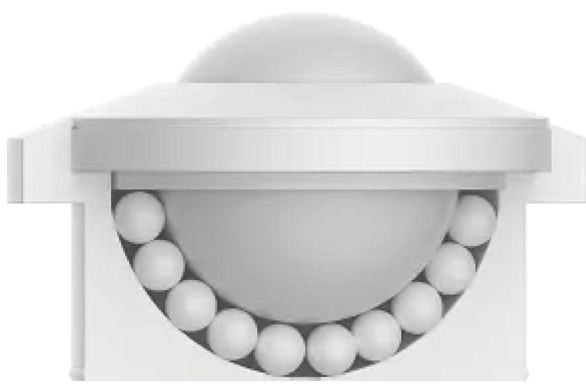

Figure 4: A ball transfer unit has many low-friction small balls surrounding the main ball [14]. *This lets the main ball spin freely while attached to the arm support platform.*

6.2 Roller Support Key Design Decisions

The reasoning behind three major design decisions for the roller support will be explained here.

1. **The support ball will be made of polystyrene.**
   Polystyrene has a surface energy of 34 dynes/cm, which is lower than the minimum of 38 dynes/cm required for inks to adhere to it [15], [16]. Therefore, it will not pick up marker ink

from the whiteboard as it rolls. Plastics like polystyrene are also very lightweight compared to alternatives like metal [17], which is better for Evaluation Criteria 3.2. Its coefficient of static friction ($\mu_s = 0.3$) is also great enough to roll without slipping (μ_s minimum is 0.12 according to [Appendix C] calculations) compared to alternatives like Teflon ($\mu_s = 0.04$) [18].

2. **The design has a symmetric wrist cuff.**
   As stated in stakeholder section 4.2, there are right-handed Engsci students that write leftward scripts who face the same issues as left-handed students writing rightward scripts. To accommodate both groups of stakeholders, the design should be ambidextrous. A wrist cuff design (Figure 5) to attach the ball to the wrist is ambidextrous as opposed to alternatives such as a wrist brace (Figure 6).

Figure 5: Wrist cuff concept that allows for an ambidextrous design.

Figure 6: Wrist brace concept that does not allow for an ambidextrous design due to asymmetry in the thumb hole placement.

3. **The wrist cuff has a velcro strap.**
   A velcro strap allows the design to fit more wrist sizes compared to alternative clasp methods (Figure 7, Figure 8). This is important as not all EngSci students have the same wrist size.

Figure 7: Two parts of a button that would be attached to either side of the wrist cuff.

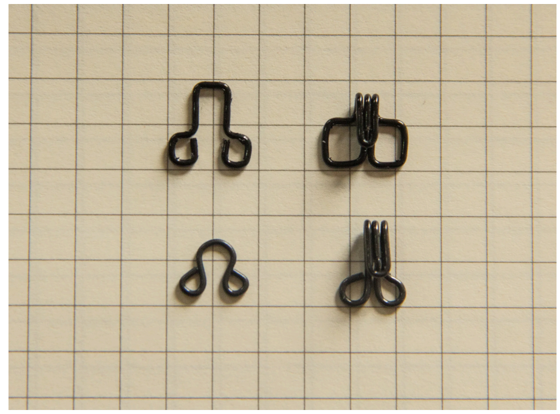

Figure 8: Clasps of different shapes that could be used to secure wrist strap.

For the button or clasp method to accommodate for varying wrist sizes, multiple buttons or clasps would have to be added. Each of this style of button tends to weigh around 2 grams, worsening the design according to evaluation criteria 3.2. Thus velcro would be the lightest solution to provide versatility for EngSci students [23].

6.3 Summary of Design Selection Rationale

The roller support design is the best solution for stakeholders because it maintains baseline writing speed and doesn't smudge marker ink while being ergonomic and lightweight (Section 8.1). When compared to other designs through Pugh Charts and Pairwise Comparison Matrices (Section 8.3) using data collected from various proxy tests, the roller support is found to perform better in most evaluation criteria.

7 Overview of Three Alternative Designs

This section will list three alternative designs to meet stakeholder needs and explain how each one functions.

7.1 Dummy Marker

Consists of a "dummy" marker held by the user which cannot draw on the whiteboard, and a second marker attached to the left of the dummy marker that actually can draw, as shown in Figure 9 and in this video. Since the ink is written on the whiteboard to the left of the hand no smudging can occur.

7.2 Air Cushion

Takes inspiration from pucks hovering over the surface of an air hockey table. This design(see Figure 10 has a platform attached to the user's wrist with a powerful fan inside that exerts a thrust on the whiteboard through a porous bottom. This provides support to the user's arm, but the surface is floating so no smudging can take place.

Figure 9: Prototype of dummy marker on a whiteboard. The marker that the user holds is taped on the bottom so that it cannot write, and the marker to the left is able to write.

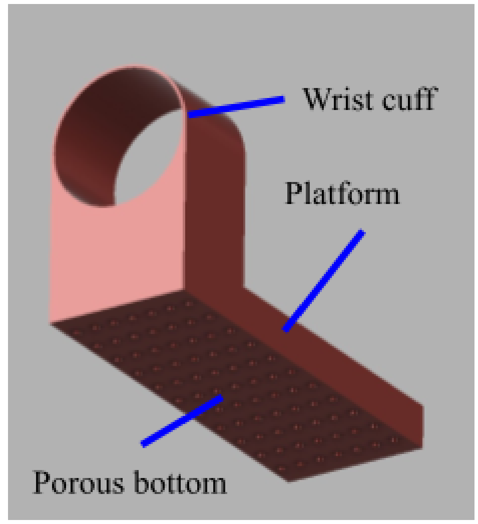

Figure 10: CAD model of air cushion. The platform contains a fan that pushes air through the porous bottom to prevent contact with the whiteboard.

7.3 Platform

A platform is attached to the marker to create a surface that creates a sense of support similar to resting their hand on the whiteboard while writing (see Figure 11). Since only the marker contacts the board, no incidental smudging will occur.

Figure 11: Platform design prototype; the user's hand rests on the attached platform while writing. The platform would be see-through.

8 Converging Process

This section will summarize each method we used in our converging process and use them to justify our recommended design and other design choices.

8.1 Verification Through Testing

This section explains each proxy test carried out, and presents results used to verify and compare designs for each requirement.

8.1.1 Proxy Test to Measure Writing Speed (Verify Requirement 1)

This proxy test for the DASH-2 assessment in Requirement 1 has a left-handed writer write a pangram with all the physical prototypes, timing how long the writer takes to write the sentence with each one. Results are shown in Figure 12, and a full procedure for this test can be found in Appendix D.

The proxy test finds that the roller support does not hinder writing speed noticeably. Writing speeds for the roller were found to be 0.99 standard deviations below the baseline writing speed, which passes Requirement 1.

On the other hand, the platform and dummy marker reduce writing speeds significantly. They reduced writing speed by more than 15 standard deviations below baseline writing speed, which does not pass Requirement 1.

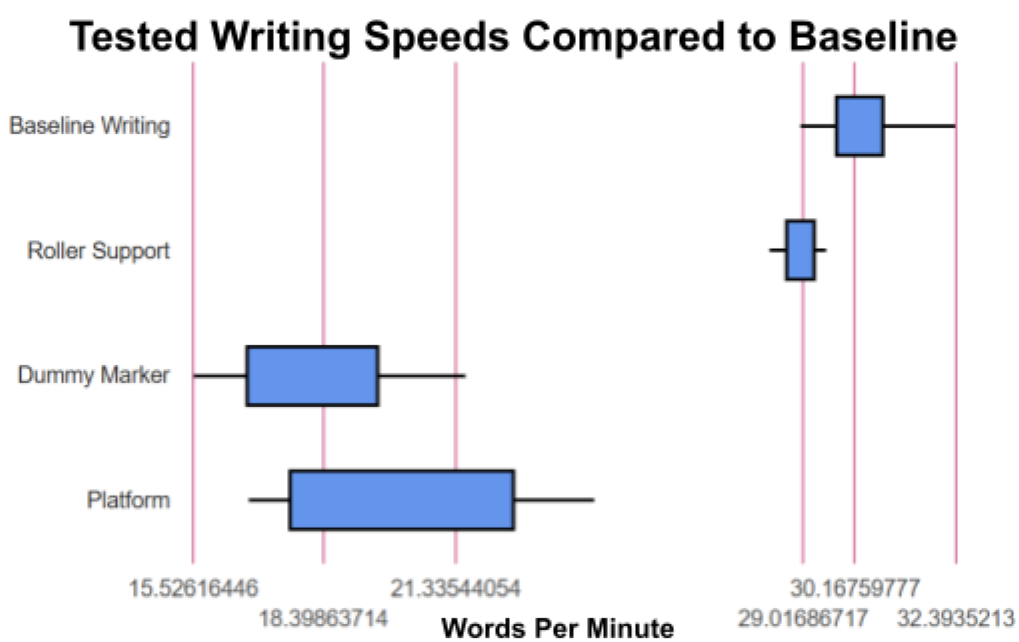

Figure 12: Box and whisker plot displaying the average writing speed using different prototypes. The platform and dummy marker significantly reduce writing speeds, because their range does not overlap the baseline speed range.

8.1.2 Proxy Test to Measure Smudging Euclidean Difference (Verify Requirement 2)

This test is only needed for the roller design because it contacts the ink. Other designs have no contact points with the ink and therefore cannot smudge, creating a ΔE* of 0.

The roller ball passes Requirement 2, meaning that its smudges are not perceivable. For this test, a roller ball was moved along the surface of some written text and an image was taken of the region, as seen in Figure 13. Then, using photoshop's color picker and a MATLAB script [Appendix D], the Euclidean difference was found to be 2.95 (Figure 14), which is lower than the maximum ΔE* = 3 in Requirement 2.

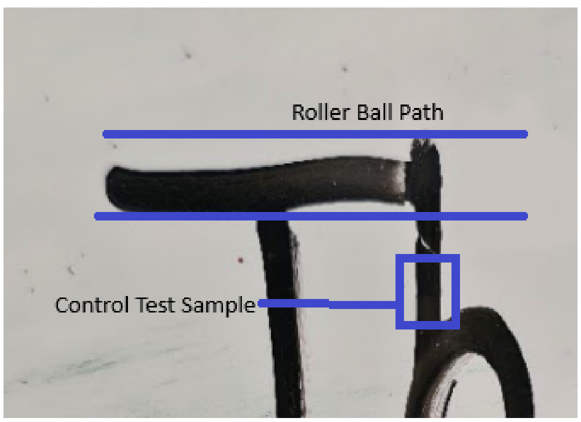

Figure 13: Shows where the color samples for the Euclidean Difference graph were taken from.

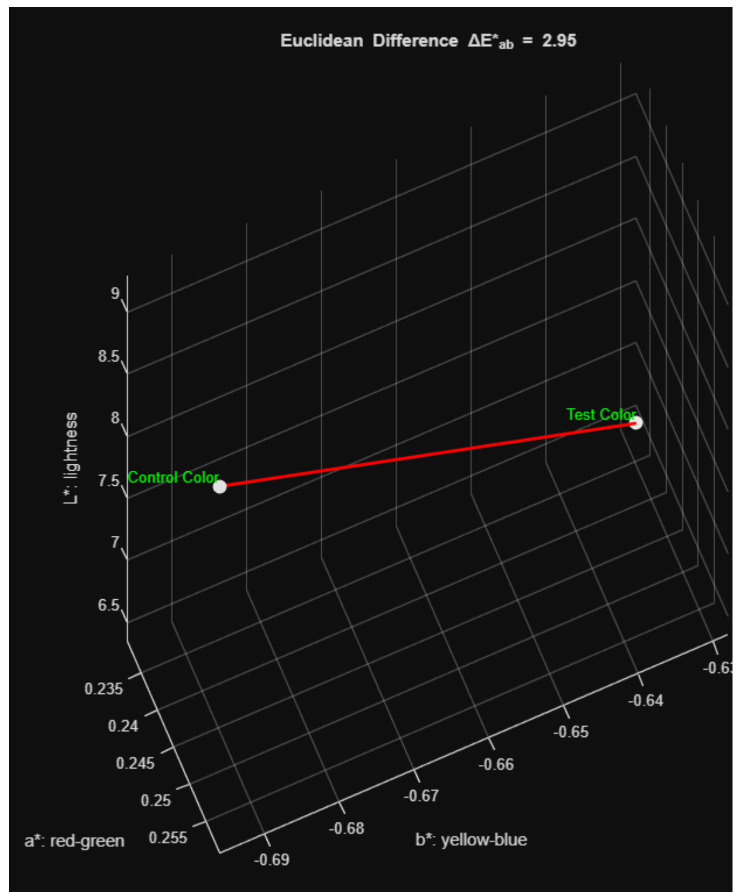

Figure 14: Position of unsmudged marker ink as a control color and position of ink color of where the roller ball passed over on the CIELAB color space, showing the ΔE.*

8.1.3 Proxy Test to Measure Wrist Angle Ergonomics (Verify Requirement 3.1)

This proxy test uses a modified version of the RULA Assessment as defined in Appendix D3. The following are the angles measured for ulnar/radial deviations and wrist extension and flexion:

Table 4: Goal and associated requirements and evaluation criteria to ensure stakeholder ergonomics and comfort.

	Ulnar/Radial Deviation	Wrist Extension/Flexion	Score	Pass/Fail Requirement 3.1
Control	17.6º	29.6º	4	Fail
Platform	20º	22.6º	4	Fail
Dummy Marker	14.3º	13.7º	2	Pass
Roller Support/Air Cushion	13.3º	14.8º	2	Pass

The dummy marker and roller support both pass the ergonomics proxy test with scores of 2. According to the RULA Assessment, any scores of 1 or 2 indicate an acceptable posture as long as it is not kept for long periods of time, meaning breaks are taken [22].

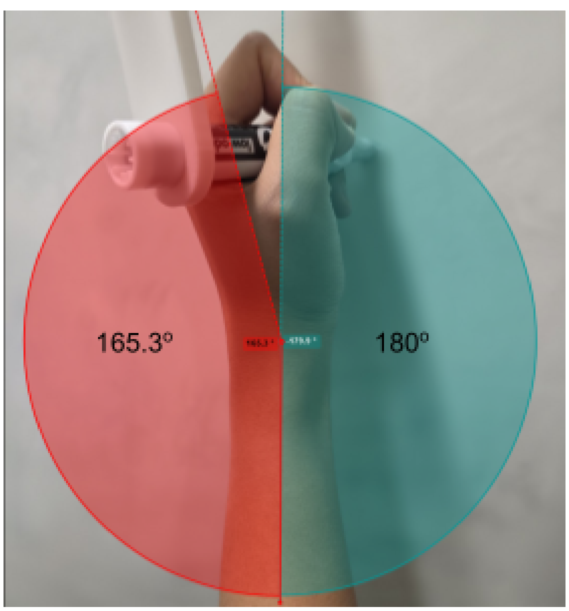

Figure 15: Example wrist extension angle measurement to measure wrist angles of dummy marker.

8.1.4 Proxy Test to Measure Design Weight (Verify Requirement 3.2)

This proxy test involved using a scale to measure prototypes. For the roller support and air cushion, estimated weights were used.

Table 5: Weights of design concepts for reference to Requirement 3.2.

	Weight In Grams	Pass/Fail Requirement 3.2
Platform	92 grams = 76 (acrylic sheet) + 16 (marker)	Pass
Dummy Marker	35 grams = 16 (marker) + 14 (dummy marker) + 5 (marker connector)	Pass
Roller Support	307.2 grams [Appendix C]	Pass
Air Cushion	578 grams = 478 (Fan & Battery) + 100 (Housing) [Appendix E]	Fail

8.2 Elimination of Air Cushion Using Pairwise Comparison Matrix

The air cushion design is not viable because it is too heavy (Section 8.1.3) and complex compared to alternatives. The air cushion row in Table 5 is all zeroes, indicating it performed worse than all other designs when compared holistically. In every comparison, the air cushion's internal fan added unnecessary weight and complexity when compared to the simpler alternatives that could provide support without smudging.

Table 6: A Pairwise Comparison Matrix comparing all designs. A 0 means design in row loses to design in column. 1 means design in row wins against design in column.

	Air Cushion	Platform	Roller Support	Dummy Marker
Air Cushion	-	**0**	**0**	**0**
Platform	**1**	-	**0**	**0**
Roller Support	**1**	**1**	-	**1**
Dummy Marker	**1**	**1**	**0**	-

8.3 Pugh Charts

Three Pugh charts will be presented to compare the designs after elimination of the air cushion. This section uses them to justify why the roller support was the final recommendation.
+ signs on Pugh Charts mean better than reference, - signs mean worse than reference, and 0 means equal to reference.

The dummy marker meets stakeholder needs better than the platform design. Table 7 shows a Pugh chart with the dummy marker as the reference. It shows that compared to the platform design, the dummy marker has faster writing speed (see Figure 12), less smudging, a smaller wrist angle (see Table 4), and weighs less.

Table 7: Pugh Chart with the Dummy Marker as Reference.

	Platform	Roller	**Dummy Marker**
Writing Speed	-	+	0
Smudge ΔE*	0	-	0
Wrist Angle	-	0	0
Weight	-	-	0

The roller design is better suited for stakeholder needs than the platform. Table 8 shows a Pugh chart with the platform as the reference design. It shows that the roller allows for faster writing speeds (Figure 12), and allows for better wrist angles (Table 4). Since smudging from the roller ΔE* is not perceivable (Section 8.1.2) and the roller weight passes verification (Section 8.1.4), the roller design is better.

Table 8: Pugh chart with the platform as the reference.

	Platform	Roller	Dummy Marker
Writing Speed	0	+	-
Smudge ΔE*	0	-	0

Wrist Angle	0	+	+
Weight	0	-	+

The roller design meets stakeholder needs better than the dummy marker and platform. Table 9 and Figure 12 shows that the dummy marker and platform have significantly slower writing speeds. Although these alternatives outperform the roller in smudging and weight in Table 9, these minor advantages do not outweigh the major writing speed drop.

Table 9: Pugh chart with the roller support as the reference.

	Platform	**Roller**	Dummy Marker
Writing Speed	-	0	-
Smudge ΔE*	+	0	+
Wrist Angle	-	0	0
Weight	+	0	+

9 Conclusion

The roller support wrist cuff is the final recommended design to let left-handed users write smudge free on whiteboards in the EngSci common room without straining their wrist in unergonomic ways. The roller design maintains baseline writing speed, minimizes visible smudging, and supports ergonomic wrist angles while remaining lightweight. When compared to other designs using Pairwise Comparison Matrices and Pugh Charts, the roller support significantly out-performs them in writing speed and only performs marginally worse in other evaluation criteria, so it is the better design. Moving forward with this design should involve testing a wide variety of materials to create an appropriate support ball that optimizes for lower weights, thus maximizing writing speed.

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11 Appendices

Appendix A: Source Extracts

[1]

English is the language of instruction and examination at the University of Toronto, and success in our programs requires a high level of English language proficiency. If an applicant's first language is not English, proof of English facility must be presented. For information on acceptable tests/qualifications, minimum scores and exemptions, please review our Discover Engineering website on English Language Requirements.

Used to find the primary language of instruction at UofT.

[2]

10.64±5.03% in right-handers. In addition, the UT muscle activation on the writing hand side was 11.91±5.79% in left-handers and 1.66±1.19% in right-handers. [Conclusion] As a result of this study, it was discovered that left-handers used more wrist flexion in performance of the writing task with the dominant upper extremity than right-handers, and that the left-handers activated the wrist and shoulder muscles more than the right-handers. These results indicate a potential danger of musculoskeletal disease in left-hander.

Links wrist flexion to musculoskeletal disorders.

[3]

Conclusion The prevalence of wrist disorders is significantly linked to wrist angles in deviation and to forces exerted. Due to their high correlation with force, the repetitiveness indices and velocities, as defined, do not appear to play an additional role. Further research is needed to find alternative ways of characterizing repetitiveness.

Links wrist flexion to musculoskeletal disorders.

[7]

1. **Surface Texture**: Porcelain whiteboards have a smooth, glass-like finish. Run your hand over the surface; it should feel sleek and resistant to scratching. If the surface feels rough or porous, it may not be porcelain.
2. **Durability**: Porcelain boards are known for their durability and resistance to wear. If the board has been used for a long time and shows minimal signs of scratching or ghosting, it is likely porcelain. In contrast, other boards, like melamine or laminate, may show signs of wear more quickly.
3. **Magnetic Properties**: Many porcelain whiteboards are magnetic, allowing you to attach notes or documents with magnets. If a magnet sticks firmly to the surface, it could indicate a porcelain board. However, note that not all porcelain boards are magnetic, so this is just one indicator.
4. **Weight**: Porcelain boards are generally heavier than their melamine or laminate counterparts. If you're able to lift the board, the weight can be a clue. If it feels particularly hefty, it's likely made of porcelain.
5. **Color and Shine**: Porcelain boards typically have a bright, glossy finish that reflects light. If the surface has a dull appearance or shows signs of fading, it may not be porcelain.

Used to identify characteristics of porcelain whiteboards.

[9]

When it comes to fine and ultra-fine tipped markers, **both Expo wet and dry erase versions are around 5.6 inches.** This variant has a more slender body akin to a pen. Let me also add that it comes in a click marker format, which appears slightly longer due to the button.

Used to find length of Expo marker

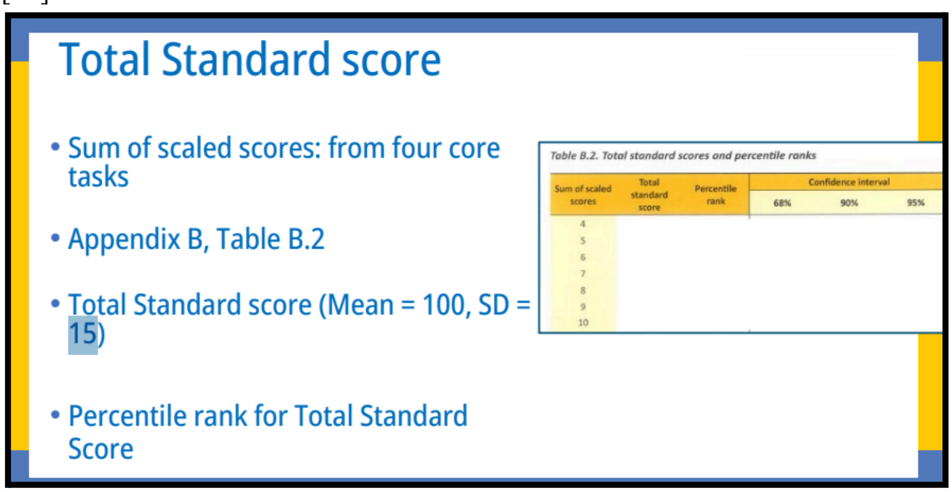
[10]

III. **Other Charges**

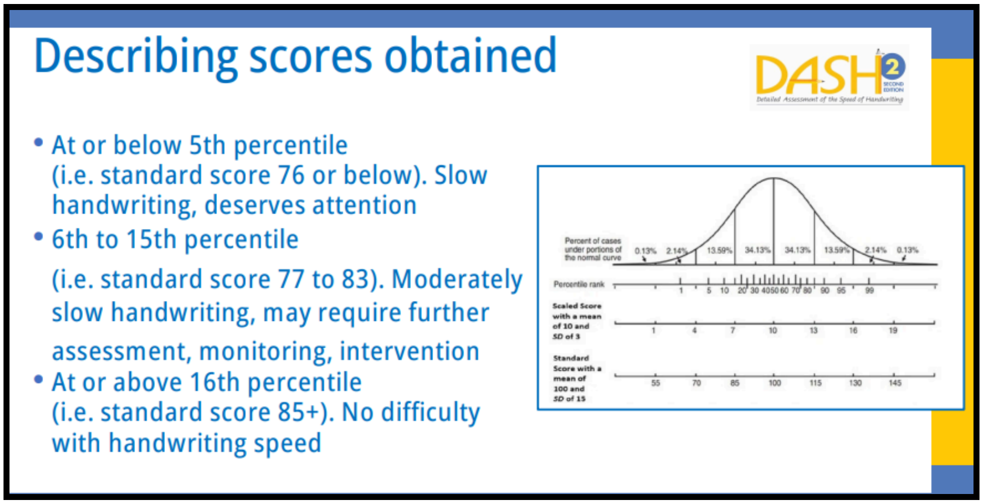
1. The University may, as a condition of bookin
where the building would normally be closec
arranged by the University.
2. The University at its discretion may assess a
normally the responsibility of the group bool
3. Over and above the rental charge and secur
   a. Use of public address, audio-visual or oth
   b. Additional caretaking costs or extraordina
   c. Special arrangements with parking and gr
   d. Special setups where applicable; and/or
   e. Damage or undue wear and tear.

Identifies acceptable conduct for students at the U of T campus

[12]



Defines in short what the DASH-2 test is with its score sheet, inspiring proxy test 1



Describes the DASH-2 score that correlates with significant reductions in handwriting. One standard deviation is found to be 15, so one standard deviation lower than baseline writing speed is significant reductions in handwriting.

[13]

Ideally, a worker should be able to operate a tool with one hand. The weight of the tool may depend on the use:

- 2.3 kg (5 lb) if the hand tool will be used away from the body or above shoulder height.
- 1.4 kg (3 lb) or less for tools operated with one hand
- 0.4kg (1 lb) for precision tools to allow for good control.

Used to justify the requirement for maximum weight of designs.

[14]

A ball transfer unit features a central ball bearing enclosed within a tubular casing, which is partially restricted. This restriction securely holds the ball bearing in place while allowing it to rotate freely, with a portion extending beyond the casing. Beneath the central ball, a layer of smaller balls is arranged in a hemispherical cup, ensuring smooth and low-friction movement in all directions.

Describes what a ball transfer unit is and how it works.

[15]

Polymer Abbr.	Polymer Name	Surface Energy (dynes/cm)	Contact Angles (degrees)
PS	Polystyrene	34	72
	Nylon-12	36	
	Surlyn ionomer	33	80

Describes polystyrene's surface energy, which is a measure of how well it attracts liquids.

[16]

Experienced printers of plastic substrates understand that the dyne level must be between 38 and 50, with 40 thought of as ideal in order to expect acceptable printing results. If the dyne level is below 38 the inks won't adhere. At the other end, if the dyne level is over 50 there will be handling problems relating to static electricity. Plastic substrates with a dyne level measured below 38 may be printed acceptably, but only after adequate surface treatment.

Describes threshold for materials that attract ink.

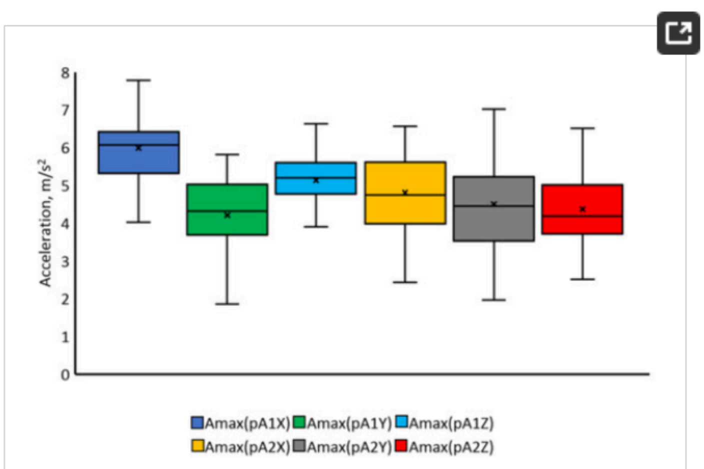
[19]

Measurement index	Writing pressure task: Strong	Writing pressure task: Weak	Writing speed task: Fast	Writing speed task: Careful
Pressure				
Pen pressure (g)	433 ± 92	83 ± 33**	222 ± 74	200 ± 63*
Writing surface pressure (g)	506 ± 114	83 ± 40**	248 ± 87	223 ± 76*
Hand pressure (g)	787 ± 259	350 ± 150**	432 ± 179	451 ± 176
Forearm pressure (g)	1,881 ± 730	1,888 ± 726	1,747 ± 639	1,815 ± 761
Degree of variation				
Pen pressure (ΔPi)	5,050 ± 830	4,060 ± 330**	5,040 ± 740	4,170 ± 450**
Writing surface pressure (ΔPi)	10,760 ± 3,100	9,220 ± 700**	10,690 ± 1,420	9,460 ± 1,200**
Writing speed (mm/s)	11.49 ± 0.31	11.70 ± 0.32**	21.85 ± 1.16	7.78 ± 0.12**

Used to find the force exerted by the hand and forearm onto the whiteboard. Note that the table says pressure (g), but this is actually a measurement of force in the units of grams-force, where 1 g = 0.0098 N.

[20]

The summary of the values of maximal accelerations and RMS (Root Mean Square) accelerations obtained during the study of reaching hand movements during the acceleration phase and the deceleration phase is shown in **Figure 13**, **Figure 14**, **Figure 15** and **Figure 16**.



Finds maximum hand acceleration during reaching hand movements that helps with calculation Appendix C.
[21]

Measurements

Item Weight	272 g

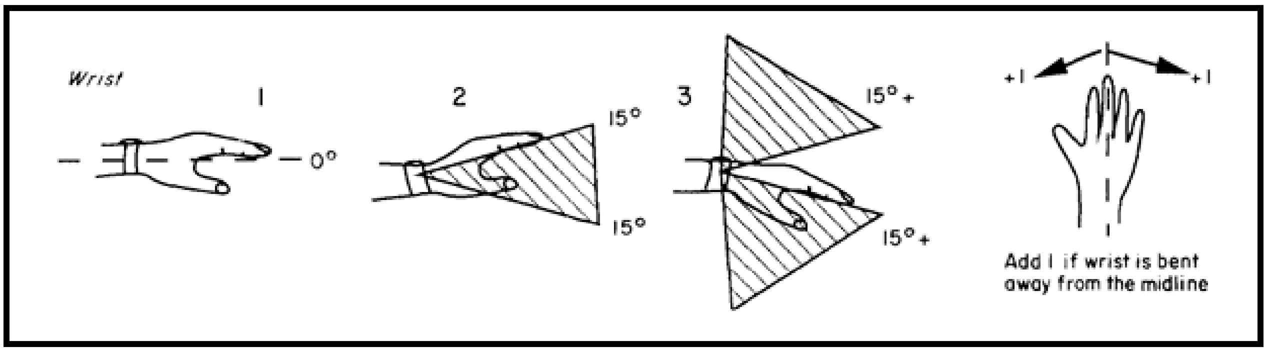
Estimates weight of wrist cuff and platform for roller support design.
[22] [Page 92]

The guidelines for the *wrist* issued by the Health and Safety Executive[23] are used to produce the following posture scores:

- 1 if in a neutral position;
- 2 for 0–15° in either flexion or extension;
- 3 for 15° or more in either flexion or extension.

If the wrist is in either radial or ulnar deviation then the posture score is increased by 1.

Defines RULA assessment.
[22] [Page 93]



Shows the extension/flexion of the wrist and ulnar/radial deviation of the wrist
[22] [Page 96]

Give a score of 1 if the posture is :

- mainly static, eg held for longer than 1 min
- repeated more than 4 times / min

Figure 4 The muscle use scores which are added to posture score A and B

0	1	2	3
• No resistance or less than 2 kg intermittent load or force	• 2- 10 kg intermittent load or force	• 2-10kg static load • 2-10kg repeated load or force	• 10kg or more static load • 10kg or more repeated loads or forces • Shock or forces with a rapid build-up

Figure 5 The force or load score which is added to posture score A and B

Defines the scoring for muscle use and force/load
[22] [Page 96]

Action level 1
A score of 1 or 2 indicates that posture is acceptable if it is not maintained or repeated for long periods.
Action level 2
A score of 3 or 4 indicates that further investigation is needed and changes may be required.
Action level 3
A score of 5 or 6 indicates that investigation and changes are required soon.
Action level 4
A score of 7 indicates that investigation and changes are required immediately.

Shows the urgency of fixing issues with ergonomics depending on points that are collected for a certain posture
[23]

Material: **Stainless steel (AISI 304)**
Surface finish: **Polished**
Weight: **2 g**

Defines the mass of buttons.
[24]

Table 1.

National bureau of standards rating [23]

National bureau of standard units	Definition of color difference
0.0-0.5	Trace (extremely slight change)
0.5–1.5	Slight (slight change)
1.5-3.0	Noticeable (perceivable)
3.0–6.0	Appreciable (marked change)
6.0–12.0	Much (extremely marked change)
12.0 and more	Very much (change to another color)

Open in a new tab

Defines noticeable ΔE* values for human sight of smudged ink.

Appendix B: Stakeholder Interviews

B1: Engineering Science Student Left-Hander 1

[Interviewer]: When you write on whiteboards, does smudging of the writing ever happen? If so, what do you do to avoid it?

[Anonymous]: "Yes – but what I do is lowkey hard to explain. I use my pinky to prop my hand up. That causes my pinky to be kinda stuck in position and causing discomfort if I write for a while."

[Interviewer]: Do you think this issue is something that bothers you often?

[Anonymous]: "Not as much as pen and paper, because the ink doesn't dry fast enough most of the time. Whiteboard is still an issue though. For writing on paper, I used these "magic" pens that don't smudge, so that idea for whiteboards sounds cool."

B2: Engineering Science Student Left-Hander 2

[Interviewer]: Could you explain some of the challenges you face in general as a left-handed individual that are not experienced by right-handers?

[Anonymous]: "Almost everything is designed for right-handed people. Like things you as a rightie have never even thought of. I'm talking about doors, scissors, binders (you know the ones with the coils)…also I always have to write like [displays ocular occlusion from position of left hand writing] to see what I'm writing. Which sucks if you have just finished writing an idea and you end up smudging all of it. Especially during [references an exam with only pens]. I got ink all over my face and I didn't realize it until after."

[Interviewer]: We are focusing our attention on whiteboards. Could you tell us about your experience with writing on whiteboards as a left-handed individual?

[Anonymous]: "Every time I go to write something, I end up messing up what I have already written. I'm mostly fine when using portable whiteboards, like the one I use as my calendar with tape and stuff, but I don't really use any wall whiteboards. So just let me think here. I think it is just a super uncomfortable experience because you almost have to contort your body and your hand so that you aren't smudging, and my wrist gets sore pretty quickly."

[Interviewer]: So what would you say is the biggest challenge if you had to pick one?

[Anonymous]: "Accidentally erasing stuff I have written, getting whiteboard markers on my hands and face (though that doesn't happen too frequently), and like the awkward writing contorting [mimes increased flexion]. This is something that no-one ever talks about, but is honestly so not inclusive for [left-handed people]."

B3: Right-Hander EngSci Student who has written in Arabic script

[Interviewer]: Do you typically face issues when writing on whiteboards in English?

[Anonymous]: "No, I usually don't face any issue when writing on whiteboards. If I'm being picky, it's annoying when whiteboards sometimes don't have an eraser on hand and I end up erasing the writing with the side of my hand because it's inconvenient to go and find one."

[Interviewer]: You have experience writing Arabic before. Did you ever face challenges writing Arabic on vertical whiteboards?

[Anonymous]: "For sure. In school, whenever I had to answer Arabic questions in front of the class on whiteboards, I'd have to angle my hand away from the board so that my palm wouldn't ruin what I was writing. I never really thought about how I don't have to do that for English, but I always felt like my handwriting on whiteboards for Arabic was much worse than on paper. Because I always angled my hand when writing, my sentences would end up completely crooked. By the time I finished a line, I'd realise the whole thing was slanted – like I'd started with the first word of a sentence at the top of the whiteboard, and somehow the last word was halfway down."

[Interviewer]: What do you mean by "ruin" your writing?

[Anonymous]: "I didn't want my writing to get smudged or erased, and I didn't want to have to wash markers off my hands after every lesson because of that either."

[Interviewer]: Do you use the EngSci common room whiteboards now at U of T?

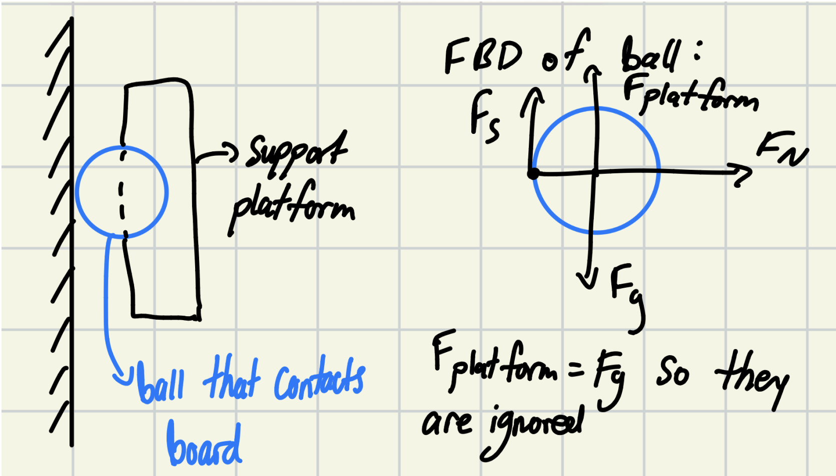
[Anonymous]: "Rarely. I don't really go to the common room much, it's really crowded whenever I've tried to go and my friends and I can never find enough seats."

[Interviewer]: Would you use a design to prevent whiteboard pen smudging?

[Anonymous]: "If I was still writing Arabic often, and something made writing Arabic on whiteboards as easy as writing English – then yes.

Appendix C: Roller Support Required Static Coefficient of Friction

The roller support only works if the ball rolls without slipping, so this appendix calculates the required coefficient of static friction for a sphere that allows it to not slip under normal handwriting conditions. This is a mathematical prototype for the roller support.



Let F_s be the force of static friction, F_N be the normal force.
$F_s = \mu_s F_N$
$\mu_s = F_s/F_N$

F_N is equal to the downwards force that the user's hand or arm will apply onto the ball. [19] finds that the minimum normal force is 350 gf + 1747 gf = 2097 gf = 20.5506 N, where 1 gf = 0.0098 N. A minimum is taken to provide a safe overestimate for μ_s, since having a lower μ_s could cause smudging.

F_s is the maximum amount of static friction force the whiteboard will have to apply onto the ball in order to move it without slipping. The maximum required friction force happens when the acceleration of the user's arm is at a maximum. According to [20], the hand usually reaches accelerations of 7.81 m/s^2.

Weight Calculation:
The weight of the design is around (272 g for the wrist cuff + $4/3*(2\ cm)^3*\pi * 1.05\ g / cm^3$ for the ball) = 307.2g = 0.3072 kg [21], [17]. This assumes the ball is 2 cm in radius and has a density similar to plastic. The wrist cuff reference is assumed to cover the weight of the platform as well, because it is a very thick wrist cuff.

Then plug into $F_s = ma = 0.3072\ kg * 7.81\ m/s^2 = 2.4\ N$.

Finally, solve for $\mu_s = F_s/F_N = 2.4\ N / 20.5506\ N = 0.12$

Therefore, in order for a sphere to not slip in usage with the roller ball support design, it must have a minimum μ_s of 0.12 against the whiteboard.

Appendix D: Proxy Test Procedures and Raw Data

Proxy Test D1: Impact of Prototypes on Writing Speed
This proxy verifies requirement 1 (design cannot inhibit user's baseline DASH-2 score) by testing writing speed with the 3 concepts, the dummy marker, the platform and the roller support. The roller support is tested by substituting for 0.4kg wrist weights to simulate the extra weight of the roller support. This

weight is an overestimate for the roller support but if it is able to pass this max threshold as defined by requirement 1, then a lighter design passes too. This substitution also works because the roller support is attached to the wrist and not to the marker, allowing the user to write in their natural position.

1. The control test defines a mean baseline writing speed for a left-handed writer writing on a whiteboard with no additional weight or using a normal marker
2. The sentence given is "The quick brown fox jumps over the lazy dog" because it contains every letter of the alphabet (pangram) and is short enough to avoid long writing session fatigue
3. Each prototype is tested, collecting a mean of 5 times to reduce error, with the 0.4 kg wrist weights used for the roller support
    a. Writing speed is considered hindered if the mean time with added weight has a z-score greater than 2 (95% confidence interval) relative to the control mean. Therefore we can establish the effect of the given weight in hindering writing speed with a significance value of a = 0.05

Proxy Test 1 Results

Table D1.1: Control-Without weights, normal writing
(9 words / 17.829 seconds) * (60 seconds / 1 minute)

Trial Number	Time (s)
1	18.45
2	17.48
3	17.93
4	17.90
5	17.76
6	17.54
7	17.80
8	18.65
9	17.90
10	16.67
11	17.18
12	18.08
13	18.30
14	18.58
15	16.98
16	17.34

17	17.90
18	18.18
19	17.76
20	18.20
Mean	**17.829**
WPM	**30.29**
Standard Deviation	**0.520**

Table D1.2: With weights (0.4kg, simulating roller ball)
(9 words / 18.33 seconds) * (60 seconds / 1 minute)

Trial Number	Time (seconds)
1	18.66
2	17.11
3	18.28
4	18.56
5	19.11
Mean	**18.344**
WPM	**29.46**
Z-score	**0.99**

Table D1.3: Styrofoam Platform
(9 words / 26.8 seconds) * (60 seconds / 1 minute)

Trial Number	Time (seconds)
1	32.26
2	25.31
3	23.81
4	22.13
5	30.65

Mean	**26.8**
WPM	**20.15**
Z-score	**17.525**

Table D1.4: Dummy marker
(9 words / 29.8 seconds) * (60 seconds / 1 minute)

Trial Number	Time (seconds)
1	29.35
2	27.47
3	25.02
4	34.78
5	32.38
Mean	**29.8**
WPM	**18.12**
Z-score	**23.02**

Proxy Test D2: Roller Smudging Euclidean Difference

This proxy test uses the Euclidean Difference MATLAB script (See Appendix x) to calculate the Euclidean difference between the hex code of the smudged and unsmudged ink.

1. Take a picture of the text with any photo processing disabled to ensure colors are unedited. This is the control and what the smudged ink will be compared to
    a. This is done with the export raw mode on the Samsung S25 Ultra
2. Then use both the roller types on the ink and take a picture of the ink
3. Input all photos into photoshop and use the color picker to get an average hex code of the color in the area that the roller was used
4. Input all the hex codes into the MATLAB script, use the outputted Euclidean Difference to determine if that roller ball material passes

Proxy Test D3: Measure Wrist Angle Ergonomics

This proxy test is based on the RULA assessment, which is a methodology developed for investigating the ergonomics of work places in such a way that no special equipment is required, allowing for a quick assessment. However, it has been adjusted to fit the need of only analyzing the wrist and not the body as a whole [22]

1. Take a picture of a left-handed writer, writing normally on a whiteboard, this will define the baseline problem

a. Two pictures are taken, one from a top down view of the lower arm and hand and one from a side view of the lower arm and hand, thus showing wrist flexion/extension and ulnar/radial deviations
b. Each picture is taken with either the hand or wrist as the central focus of the image and all angles are then relative to that
    i. Choosing any central focus is fine as there is no impact to resultant angle as it is calculated as a difference between two angles

2. The picture is uploaded to *Kinovea video analyzer*, which thus allows placing a digital goniometer on the wrist joint to measure the wrist flexion/extension and ulnar/radial deviations
3. This is then repeated with the platform and dummy marker physical prototypes and the weight wrist cuffs to simulate the roller ball support
    a. The weight wrist cuff can simulate the roller ball support because the roller ball is not connected to the actual marker so the marker grip remains the same as the normal grip used by left-handed writers when writing on a whiteboard without worrying about not smudging the text (inverse right-handed grip) and the roller ball support in theory just lift the wrist up vertically off the whiteboard but keeping the same writing angle
4. The point system defined in the RULA Assessment has been adjusted as following:
    a. Only the wrist portion of the score is considered
    b. Muscle score is given a score of 1 for all prototypes as they are all held for more than a minute and have repeated movements of writing [22]
    c. Give force score of 0 to prototypes with weight less than 0.4 kg and score of 1 to prototypes with weight greater than that
        i. Writing is done for longer periods of time in a vertical position so fatigue is felt at a much faster rate, so a lower weight is preferred and thus a lower weight threshold than the RULA assessment is recommended. Additionally, requirement 1 defines that for safe use, the design weight should not exceed 0.4 kg
    d. Wrist is given a score of 1 for flexion or extension within 15^{o} of the neutral position and 2 points for greater than 15^{o}. 1 additional point for any ulnar or radial deviation greater than 15^{o} as defined by the RULA assessment
    e. The action levels defined in the RULA assessment are not completely applicable as RULA is holistic of the whole body while this test only considers the wrist. As within 15^{o} is considered to be acceptable deviations, a score of 2 will be taken as a pass for this test (which would be 1 point for less than 15^{o} flexion or extension and 1 point for muscle score)

Appendix E: Air Cushion Spec

Thrust Output Calculation for 70 mm EDF (Turbines RC-FD Model) [25]

Specifications:
- EDF Model: JP Hobby 70 mm, 12-blade
- Motor: 3400 KV, 4S compatible
- Total weight (EDF + motor + shroud): 178 g

Battery Specifications:

- Battery type: 4S LiPo (14.8V nominal)
- Capacity: 3500 mAh
- Discharge rating: 35C continuous
- Battery weight: ~300 g
- Connector: XT60 or Deans

Total System Weight:
- EDF unit: 178 g
- Battery: 300 g
- Total propulsion system: 478 g
- Operating voltage: 4S LiPo (14.8 V nominal)
- Current draw: 75 A at full throttle

Electrical Power Calculation:
$P = V \times I$
$P = 14.8\ V \times 75\ A$
$P \approx 1110\ W$ (manufacturer spec: 1260 W)

Measured Static Thrust:
From manufacturer test data at 4S, 75 A:
$T = 1810\ g = 1.81\ kgf$

Conversion to Newtons:
$T = m \times g$
$T = 1.81\ kg \times 9.81\ m/s^2$
$T = 17.8\ N$

Summary:
The Turbines RC-FD 70 mm EDF (178 g, model FED-7012-3400) produces approximately 17.8 N (1.81 kgf) of static thrust at full throttle on a 4S LiPo battery, drawing 75 A and consuming approximately 1260 W of electrical power. This thrust output is achieved with a compact 70 mm diameter form factor and a total system weight of 178 grams including the motor, fan, and synthetic shroud. When paired with a recommended 4S 3500mAh 35C LiPo battery (~300 g) [26], the complete propulsion system weighs approximately 478 grams.

Appendix F: Original Design Brief

Page numbers for the original design brief are in the lower left hand corner.

IMPROVING ENGINEERING SCIENCE COMMON ROOM WHITEBOARD USABILITY FOR LEFT-HANDERS

Word count: **1798**

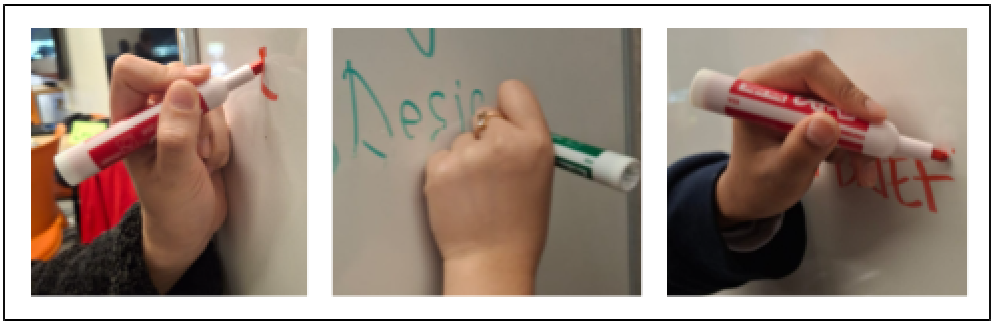
Table of Contents:

I. Introduction

This design brief frames the challenges faced by left-handed writers when using whiteboards in the Engineering Science common room. Left-handers often smudge or erase their written work unintentionally when writing rightwards [Appendix B2-3], [Appendix C]. To limit these effects, left-handers resort to uncomfortable writing postures, resulting in muscle strain and poor handwriting [3], [4], [28]. This brief defines the scope of this problem, discusses stakeholder expectations, and explores the limitations of existing designs to inform needs, goals, and objectives for improving the left-handed whiteboard experience.

II. Using EngSci Common Room Whiteboards as a Left-Handed Individual

As most language scripts in the world are right-oriented [2], left-handers' hands tend to trail over their script and rest on the board (Figure 1), smudging ink (Figure 2) and transferring residue to the palm [Appendix B3], [Appendix C]. The non-absorbent whiteboard surface causes this inconvenient writing experience. For the purpose of this design brief, the whiteboards are a fixed element of the common room that can not be replaced.



1.1 1.2 1.3

Figure 1. Observation of different left-handed pen grips used by students when using whiteboards

Observation and research shows that left-handers often adopt awkward wrist angles when writing [Figure 1.3], [Appendix B3], [Appendix C]. Even on horizontal surfaces, they exhibit greater wrist flexion than right-handers [3], increasing muscle strain and the risk of musculoskeletal disorders [3], [4], [27]. This effect is likely exacerbated on whiteboards, where the arm is unsupported and a standing stance is adopted. Awkward wrist adjustments can also reduce perceived handwriting quality [Appendix B2], [Appendix C3], [28].

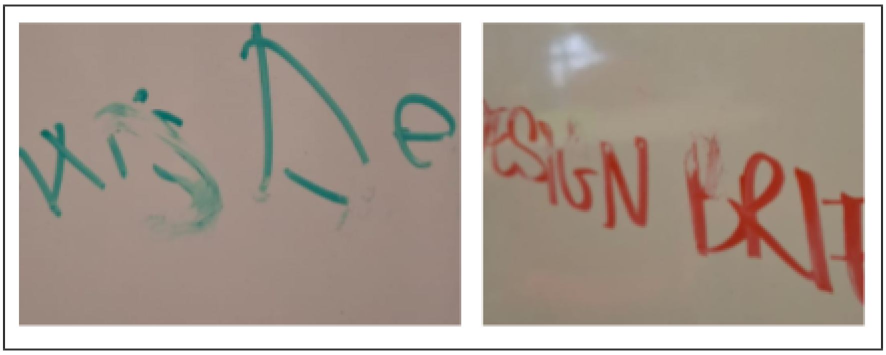


Figure 2. Observations of smudging yielded from left-handers in Fig. 1

Hence, despite posture adjustments, whiteboard writing still causes discomfort and reduced satisfaction for left-handers, emphasizing the need for a design solution.

III. Defining the Working Environment

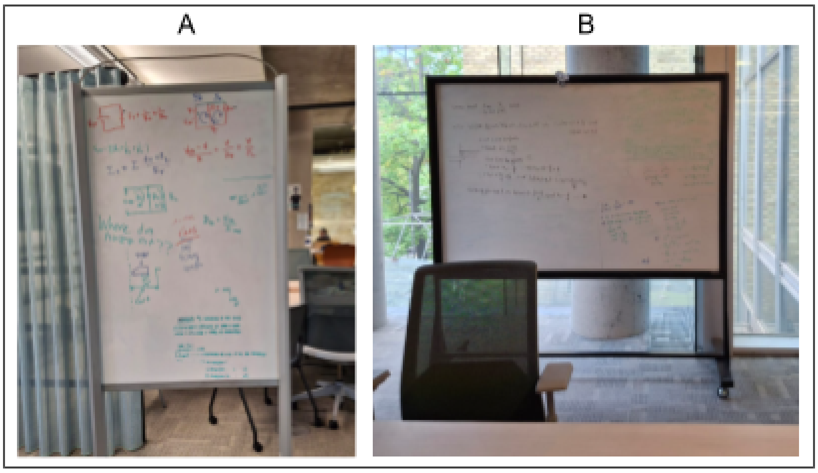


Figure 3. The 2 whiteboard models that were considered for this design brief.

The Engineering Science Common Room is an indoor space with several whiteboards, each with a tray for markers and erasers. While the board material is unconfirmed, it is likely melamine as it accounts for the majority of the whiteboard market (45%) [5].

There are 2 sizes of whiteboards in the common room; tray dimensions were found to be 5.5 ± 0.05 cm (Whiteboard A) (Figure 4) and 4.9 ± 0.05 cm (Whiteboard B). Whiteboard markers in the common room are 17.8 cm [14] in length and the eraser is 13 cm x 8 cm x 5 cm [17] . Design solutions could employ similar dimensions to maintain consistency with standard whiteboard materials, increasing users' adaptability as outlined in Norman's 'The Design for Everyday Things' [24]. Since the common room is frequently occupied by Engineering Science students studying [Appendix C3], the design must also avoid causing auditory disturbance.

Figure 4: tray of whiteboard A.

IV. Understanding Stakeholders and Their Needs

As the common room is a shared educational space, there are many who may be affected by a design solution. Below is a discussion of different stakeholders and their expectations.

a) *Left-Handed EngSci Students*

The primary stakeholders – users of this opportunity.

They expect to be able to write on whiteboards at the rate as right-handed peers without unintentional smudging of the writing and on the hand [Appendix B2]. The design should be easy and fast to implement into one's writing process without affecting their writing ability. To ensure this, it should prevent hand pain and should be ergonomic [Appendix C1].

b) *Audiences and Group Members of Left-Handed Students*

Primary stakeholders.

They expect to read the writing of the left-handed peers' work on whiteboards without issue. The design should minimize noise to avoid distraction [23]. It must be operable efficiently and quick to access, without hindering the user's working pace.

c) *Right-Handed Students Writing Leftward Scripts (e.g, Arabic)*

Potential primary stakeholders with similar struggles. [Appendix C3]

They wish to write with satisfactory handwriting and note minimal differences between writing left-ward and right-ward scripts [Appendix C3]. The design could either not consider handedness in design or have a right-handed alternative.

d) *Facility Custodians*

Secondary stakeholders

They expect that the design can not leave non-removable residue and must not require maintenance multiple times a day, as stakeholders are only on site to clean once daily [Appendix A]. Design must pass the University of Toronto APPA Cleaning Standard [29].

e) *Facility Equipment Regulator*

Tertiary stakeholders.

They expect the design to not damage the whiteboard or surrounding equipment. This design should pass any standards from the University of Toronto for educational spaces [31].

f) Design Team

Primary stakeholders.

They expect other stakeholders' expectations to be satisfied by fulfilling the Need, Goals and Objectives. This design should carry minimal bias from the team's perspective through in depth research and understanding of other stakeholders.

V. A Discussion of Reference Designs

While some existing products may aid with left-handed whiteboard use, each has shortcomings relative to stakeholder needs.

a) Integrated Eraser Markers

Permanent markers with integrated erasers (e.g., Faber-Castell Multimark [5]) offer a short-term solution: the permanent ink prevents smudging during writing, and the eraser on the back allows removal afterward. While Faber-Castell does not disclose its eraser formula, it is likely alcohol-based since permanent marker polymers are soluble in alcohols such as isopropanol [6].

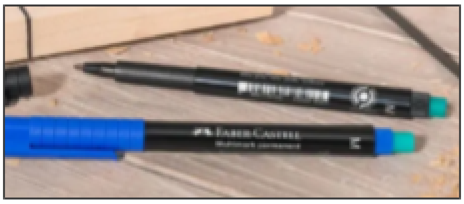


Figure 5. [5] Multimark pen with integrated eraser on back of pen, in teal.

However, the eraser's small surface area makes cleaning large sections slow. Enlarging it might seem effective, yet another issue remains: alcohol-based erasers degrade melamine whiteboards with repeated use, shortening board lifespan [8]. Hence, while integrated eraser markers address smudging, they are not a sustainable long-term solution – however, they may inspire chemical designs based on how they erase non-smudge ink.

b) Smudge Guards

SmudgeGuard® gloves (87% Nylon, 13% Spandex) [9] are intended for tablet users looking to reduce sweat transfer to screens; thus, they may also prevent ink smudging (Figure 6).

However, this design cannot fully prevent unintentional erasure since it may lift dry-erase ink from the whiteboard due to the design's absorbency. While inspiration can be taken from how they protect the palm, they highlight the need for a solution to prevent erasure.

Figure 6. [9] SmudgeGuard® gloves, which cover the palm, minimising ink transfer.

c) *InstaMorph*

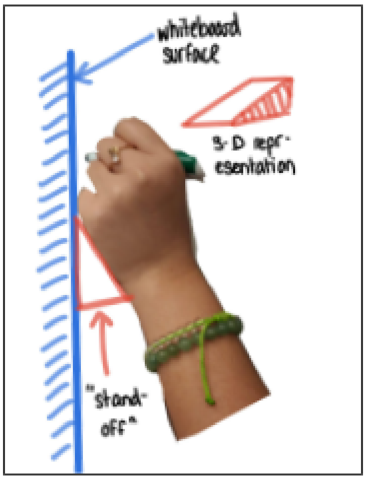


Figure 7. simple representation of what a "stand-off" might mean.

One left-hander expressing annoyance on Quora suggested using InstaMorph, forming a "custom stand-off" (Figure 7) with a lightweight polyester thermoplastic [11], to form to pivot the hand while writing on boards [Appendix B #1].

However, this task requires modeling skills and dedicated time to create a personalized product. The prolonged sensation of plastic against skin is also undesirable [21], and contact with the board risks scratches. While InstaMorph is an impractical widescale solution, the pivot concept could inspire future designs.

VI. Needs, Goals and Objectives

From the stakeholder expectations and reference design shortcomings, we can derive needs, goals and objectives necessary to deem a solution to this *splartz* "sufficient."

Need: Left-handed students are able to write on whiteboards in the EngSci common room without issue.

Table 1: Goal 1 meets primary stakeholders' needs by facilitating their use of whiteboards without harm.

	Description	Metric	Justification
Goal 1	Design must not affect the user's baseline writing ability		
Objective 1.1	Cannot inhibit the users baseline DASH-2 score by more than 15 points [16].	Score (points) on the DASH-2 (Detailed Assessment of Speed of Handwriting).	A 15 points decrease on the DASH-2 is one standard deviation, signifying a significant drop in handwriting speed [16].
Objective 1.2	Ensure there is no risk of strain injury caused by wrist extended by >90°, wrist radial abduction, >20°, and wrist ulnar abduction, >30°, outlined in ISO 11226-2000 [Appendix A] [13].	Angle of extreme wrist extension, wrist radial abduction and wrist ulnar abduction in degrees.	Reducing strain injuries when adopting a standing stance re-ensures the user's original writing ability.

Objective 1.3	Accommodate for various grip styles on the pen, namely tripod and quadropod grip styles.	Grip styles: lateral tripod, dynamic tripod, lateral quadropod, dynamic quadropod.	Ensures the design is usable by any user with differing grip styles [14]. DfErgonomics [27] emphasises to "anticipate actions."

Table 2: Ergonomics-focused goal that meets the needs of all primary stakeholders through ensuring the device is convenient to use and be around.

	Description	**Metric**	**Justification**
Goal 2	Design must be comfortable to use and have around the commons.		
Objective 2.1	Design width of at most 4.9cm.	Width (cm) of design.	Design must fit onto the marker trays (4.9cm and 5.5cm). so users can have quick access.
Objective 2.2	Design colours are distinguishable, colour difference between the design and the marker rail (silver, #C0C0C0) must have a perceptual $\Delta E^* \geq 20$ according to ISO/CIE 11664-4 [Appendix A], [12].	The Euclidean Difference (ΔE^*).	A visually distinct colour allows users to have quick access to the design.
Objective 2.4	Reduce disturbance by maintaining the common room's sound pressure level to under 60 dB.	Room's sound pressure level (dB).	Research shows that "increasing background sound pressure level to 60 dBA significantly impairs auditory working memory task performance" [22].

Table 3: Goal 3 caters to primary stakeholders. Research suggests that smudging leads to lower legibility in left-handed children [28]; hence, we assume that by reducing smudging, one increases legibility. Further research could be done to confirm.

	Description	Metric	Justification
Goal 3	Ensure no unintentional movement of the ink on the surface of the whiteboard occurs		
Objective 3.1	No accidental contact between design and surface of the whiteboard within 27.73s [19] of the ink being applied.	Ink's drying time (seconds), contact points.	Ensures no unintentional smudging can occur before the ink is dry [19], maintaining legibility.
Objective 3.2	No accidental contact between design and surface of the whiteboard after 27.73s of the ink being applied.	Ink's drying time (seconds), contact points.	Ensures no writing is erased after ink drying [19] so written work can be read.

Table 4: Goal 4 reduces the messiness of the writing experience – an inconvenience mentioned by left-handers [Appendix B, C].

	Description	Metric	Justification
Goal 4	Ensure no ink residue is left on the hand after use of the device		
Objective 4.1	Zero points of contact between the hand of the user and the surface of the whiteboard.	Hand-whiteboard contact points.	Ensures no ink is able to rub off onto the skin of the user, fulfilling stakeholder's expectation.

Table 5: Goal 5 designs for maintenance, satisfying custodians and regulatory staff through reduced damage and stains to whiteboards.

	Description	Metric	Justification
Goal 5	Design must not damage the whiteboard.		

Objective 5.1	Any nicks on the whiteboard surface must be “no greater than 0.005″ (0.01270 cm) in depth and 0.010″ (0.0254 cm) in width.”	Depth and width of nicks (inches/cm).	PTI Technologies’ Surface Inspection Acceptance Criteria allows for surface irregularities below 0.005″ in depth and 0.010″ in width on “non-functional surfaces”, which a whiteboard falls under [25].
Objective 5.2	Any staining from the design should have $\Delta E^* \leq 1.9$ with whiteboard surface.	The Euclidean Difference (ΔE^*).	Ensures color changes remain below the perceptible threshold in a dental study where 50% of participants noticed differences in dentures [26]; design must be below this to avoid visible staining.

VI. Next Steps

In conclusion, left-handed smudging on boards and palms when using EngSci common room whiteboards is a documented issue. A design team addressing this challenge must maintain regular communication with primary stakeholders to ensure that left-handers are satisfied with the final outcome. While literature is available on left-handed struggles, further primary research within the common room environment may benefit the design team. Through divergent thinking and rigorous testing to meet goals and objectives, a design team can effectively address the needs of left-handers and develop a practical solution.

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See Appendix A for relevant source extracts.

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VIII. Appendices

Appendix A: Relevant Source Extracts

[3] Page 1

1.66±1.19% in right-handers. [Conclusion] As a result of this study, it was discovered that left-handers used more wrist flexion in performance of the writing task with the dominant upper extremity than right-handers, and that the left-handers activated the wrist and shoulder muscles more than the right-handers. These results indicate a potential danger of musculoskeletal disease in left-hander.

[4] Page 1

Results All the derived variables were highly correlated, greater angles and greater forces being associated with greater velocities and higher repetitiveness. A multivariate linear regression model for the prediction of the prevalence of musculoskeletal disorders of the wrist was constructed ($R = 0.904$). Height, weight, seniority, angles in radial-ulnar deviation, and forces were significant and independent predictors of the prevalence.
Conclusions The prevalence of wrist disorders is significantly linked to wrist angles in deviation and to forces exerted. Due to their high correlation with force, the repetitiveness indices and velocities, as defined, do not appear to play an additional role. Further research is needed to find alternative ways of characterizing repetitiveness.

[6]

But what if the board gets stained? It's important to remember that the polymers in both permanent and dry erase markers are water-resistant, so the best way to clean any stubborn smears on your board is to add a little squirt of alcohol. The most effective options contain at least one of the solvents used in the markers themselves, such as isopropanol. This is found in rubbing alcohol as well as nail varnish remover and should quickly disperse the ink before it evaporates, allowing you to rub it off. Since isopropanol is found in dry erase pens, drawing over a whiteboard stain and then rubbing with a cloth can also help remove it.

[21] Page 5